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Class Spliterators.AbstractDoubleSpliterator

java.lang.Object
  java.util.Spliterators.AbstractDoubleSpliterator

All Implemented Interfaces:

Spliterator<Double>, Spliterator.OfDouble, Spliterator.OfPrimitive<Double,DoubleConsumer,Spliterator.OfDouble>

Enclosing class:

Spliterators

```
public abstract static class Spliterators.AbstractDoubleSpliterator
extends Object
implements Spliterator.OfDouble
```

An abstract `Spliterator.OfDouble` that implements `trySplit` to permit limited parallelism.

To implement a spliterator an extending class need only implement Spliterator.OfDouble.tryAdvance(java.util.function.DoubleConsumer) tryAdvance}. The extending class should override Spliterator.OfDouble.forEachRemaining(java.util.function.DoubleConsumer) forEach} if it can provide a more performant implementation.

API Note:

This class is a useful aid for creating a spliterator when it is not possible or difficult to efficiently partition elements in a manner allowing balanced parallel computation.

An alternative to using this class, that also permits limited parallelism, is to create a spliterator from an iterator (see Spliterators.spliterator(java.util.PrimitiveIterator.OfDouble, long, int). Depending on the circumstances using an iterator may be easier or more convenient than extending this class. For example, if there is

already an iterator available to use then there is no need to extend this class.

Since:

1.8

See Also:

Spliterators.spliterator(java.util.PrimitiveIterator.OfDouble, long, int)

Nested Class Summary

Nested classes/interfaces inherited from interface java.util.Spliterator

Spliterator.OfDouble, Spliterator.OfInt, Spliterator.OfLong, Spliterator.OfPrimitive<T,T_CONS,T_SPLITR extends Spliterator.OfPrimitive<T,T_CONS,T_SPLITR>>

Field Summary

Fields inherited from interface java.util.Spliterator

CONCURRENT, DISTINCT, IMMUTABLE, NONNULL, ORDERED, SIZED, SORTED, SUBSIZED

Constructor Summary

Constructors

Modifier	Constructor	Description
protected	**AbstractDoubleSpliterator**(long est, int additionalCharacteristics)	Creates a spliterator reporting the given estimated size and characteristics.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
int	characteristics()	Returns a set of characteristics of this Spliterat
long	estimateSize()	Returns an estimate of the number of elements encountered by a Spliterator.forEachRemaining(java.util.f super T>) traversal, or returns Long.MAX_VALU unknown, or too expensive to compute.
Spliterator.OfDouble	trySplit()	If this spliterator can be partitioned, returns a S elements, that will, upon return from this metho this Spliterator.

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Methods inherited from interface java.util.Spliterator.OfDouble

forEachRemaining, forEachRemaining, tryAdvance, tryAdvance

Methods inherited from interface java.util.Spliterator

getComparator, getExactSizeIfKnown, hasCharacteristics

Constructor Detail

AbstractDoubleSpliterator

```
protected AbstractDoubleSpliterator(long est,
                                    int additionalCharacteristics)
```

Creates a spliterator reporting the given estimated size and characteristics.

Parameters:

`est` - the estimated size of this spliterator if known, otherwise Long.MAX_VALUE.

`additionalCharacteristics` - properties of this spliterator's source or elements. If SIZED is reported then this spliterator will additionally report SUBSIZED.

Method Detail

trySplit

```
public Spliterator.OfDouble trySplit()
```

If this spliterator can be partitioned, returns a Spliterator covering elements, that will, upon return from this method, not be covered by this Spliterator.

If this Spliterator is `Spliterator.ORDERED`, the returned Spliterator must cover a strict prefix of the elements.

Unless this Spliterator covers an infinite number of elements, repeated calls to `trySplit()` must eventually return `null`. Upon non-null return:

- the value reported for `estimateSize()` before splitting, must, after splitting, be greater than or equal to `estimateSize()` for this and the returned Spliterator; and
- if this Spliterator is `SUBSIZED`, then `estimateSize()` for this spliterator before splitting must be equal to the sum of `estimateSize()` for this and the returned Spliterator after splitting.

This method may return `null` for any reason, including emptiness, inability to split after traversal has commenced, data structure constraints, and efficiency considerations. This implementation permits limited parallelism.

Specified by:

`trySplit` in interface `Spliterator<Double>`

Specified by:

`trySplit` in interface `Spliterator.OfDouble`

Specified by:

`trySplit` in interface `Spliterator.OfPrimitive<Double,DoubleConsumer,Spliterator.OfDouble>`

Returns:

a Spliterator covering some portion of the elements, or null if this spliterator cannot be split

estimateSize

```
public long estimateSize()
```

Returns an estimate of the number of elements that would be encountered by a `Spliterator.forEachRemaining(java.util.function.Consumer<? super T>)` traversal, or returns `Long.MAX_VALUE` if infinite, unknown, or too expensive to compute.

If this Spliterator is Spliterator.SIZED and has not yet been partially traversed or split, or this Spliterator is Spliterator.SUBSIZED and has not yet been partially traversed, this estimate must be an accurate count of elements that would be encountered by a complete traversal. Otherwise, this estimate may be arbitrarily inaccurate, but must decrease as specified across invocations of Spliterator.trySplit().

Specified by:

estimateSize in interface Spliterator<Double>

Implementation Requirements:

This implementation returns the estimated size as reported when created and, if the estimate size is known, decreases in size when split.

Returns:

the estimated size, or Long.MAX_VALUE if infinite, unknown, or too expensive to compute.

characteristics

```
public int characteristics()
```

Returns a set of characteristics of this Spliterator and its elements. The result is represented as ORed values from Spliterator.ORDERED, Spliterator.DISTINCT, Spliterator.SORTED, Spliterator.SIZED, Spliterator.NONNULL, Spliterator.IMMUTABLE, Spliterator.CONCURRENT, Spliterator.SUBSIZED. Repeated calls to characteristics() on a given spliterator, prior to or in-between calls to trySplit, should always return the same result.

If a Spliterator reports an inconsistent set of characteristics (either those returned from a single invocation or across multiple invocations), no guarantees can be made about any computation using this Spliterator.

Specified by:

characteristics in interface Spliterator<Double>

Implementation Requirements:

This implementation returns the characteristics as reported when created.

Returns:

C a representation of characteristics

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